

AuraFS: Physics-Informed Distributed Storage

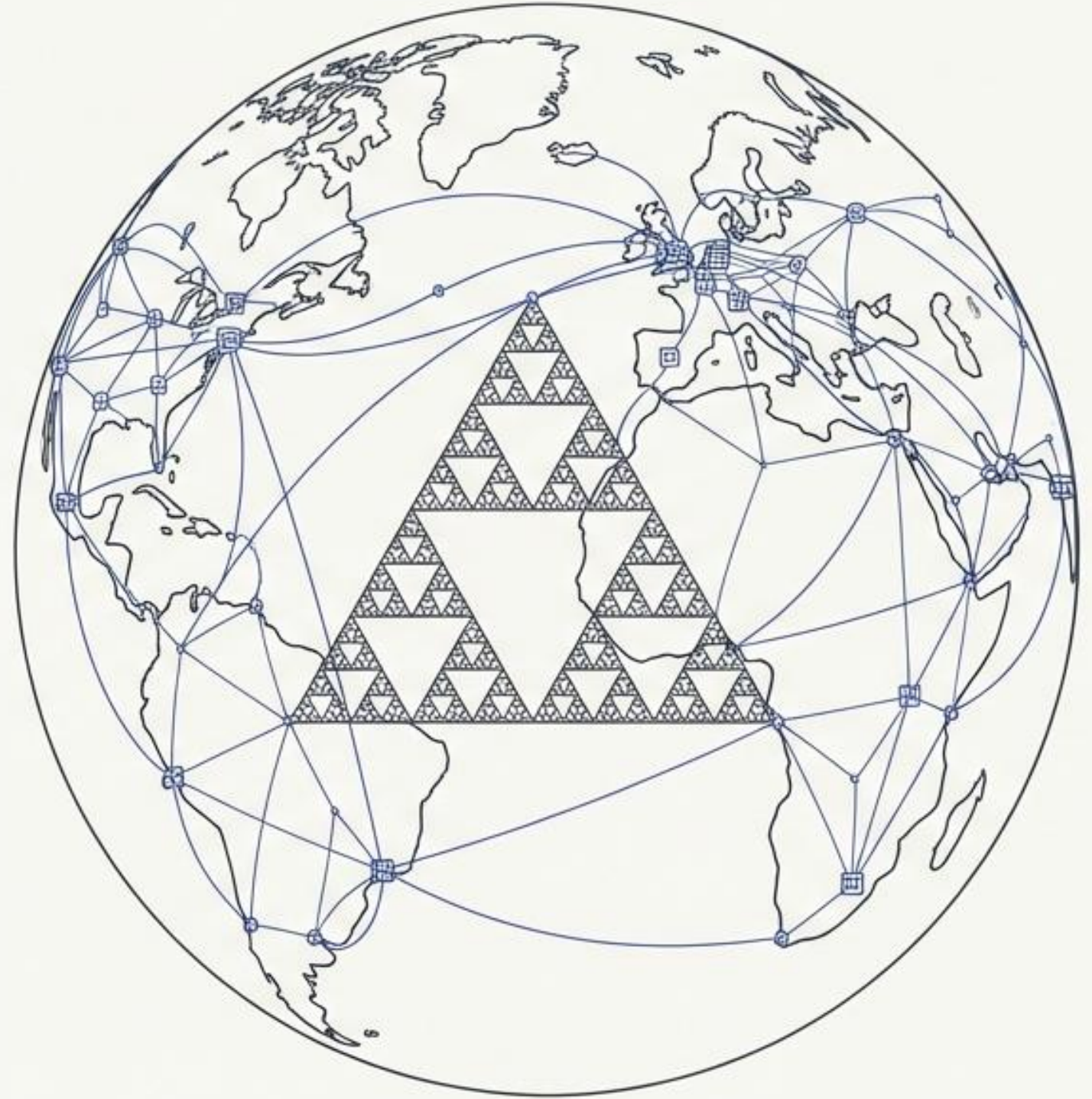
Phase II • TRL-4 Lab-Validated Architecture

Project: Aurphyx LLC

Architecture: Non-Semisimple TQFT & Anderson Localization

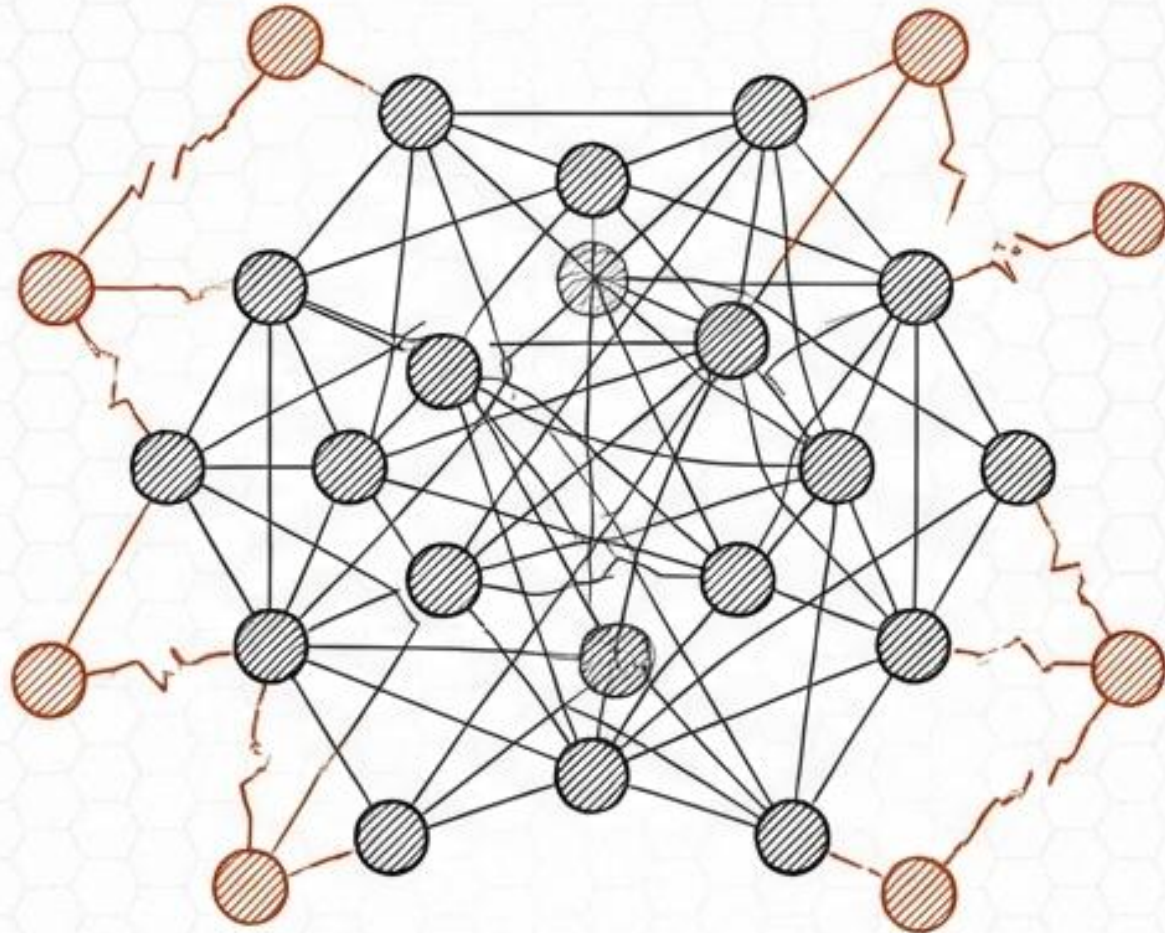
Target Readiness: TRL-4 (Lab-Validated)

Cryptographic Substrate: Post-Quantum (NIST Level 5)



Software redundancy is fragile; physics invariants are absolute.

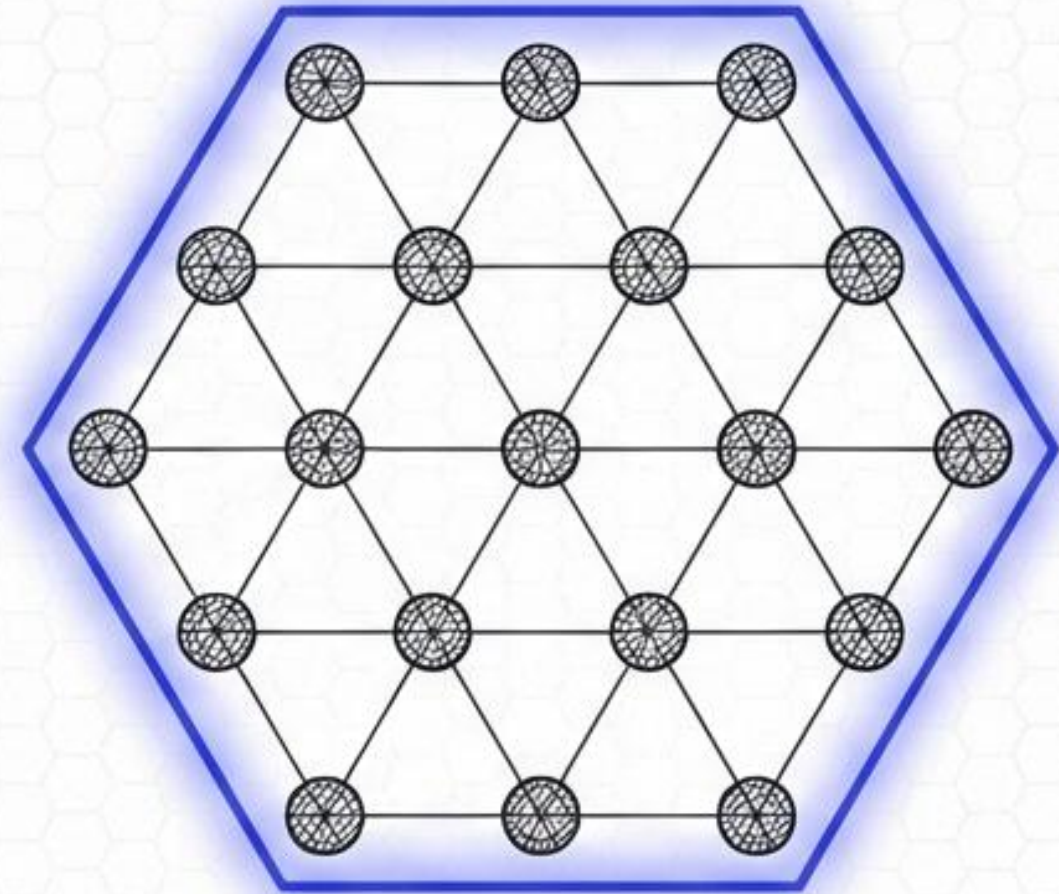
Legacy Euclidean Storage



+ The Problem

Conventional distributed filesystems rely purely on software algorithms that fail catastrophically under quantum threats or massive infrastructure collapse.

AuraFS Topology



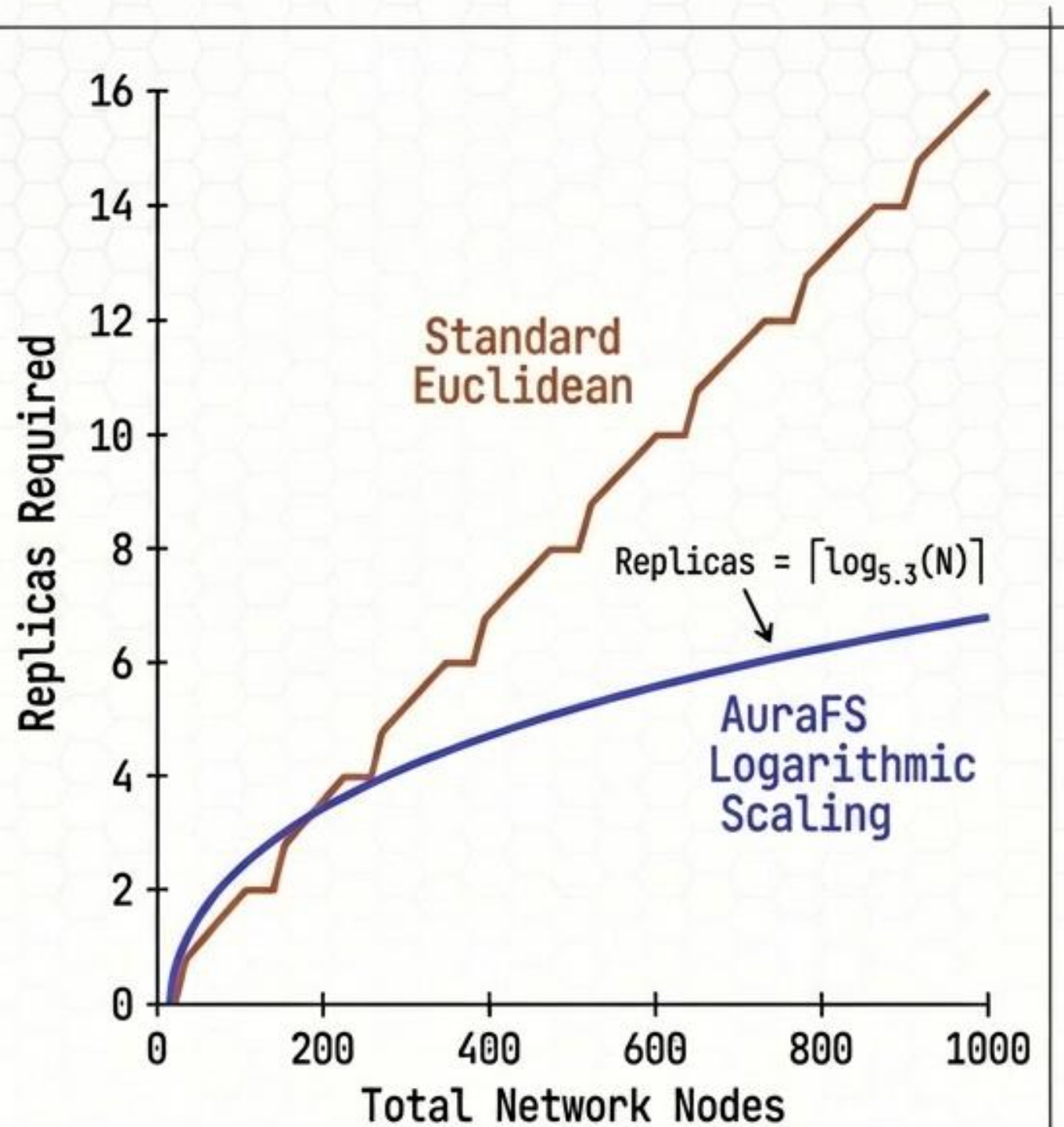
+ The Paradigm Shift

AuraFS hardcodes resilience into the storage substrate itself by leveraging physical properties of fractal geometry, photonic band gaps, and anomalous diffusion. Physical laws, not algorithms, guarantee data integrity.

The Five Sacred Constants Governing the AuraFS Substrate

Physics Substrate Diagnostic Table			
Symbol	Hard Value (Tolerance)	Physical Meaning	System Impact
η	5.3 (± 0.05)	Hilbert Scaling Bias	Determines sub-linear replica scaling requirements based on network size.
T_2	1600 μs (± 100)	Coherence Window	The maximum permitted duration for synchronous state transitions before decoherence triggers.
d_s	1.37 (± 0.05)	Spectral Dimension	Dictates anomalous diffusion. A value < 2 guarantees absolute data localization.
PBG	0.21 (± 0.03)	Photonic Band Gap	The fractional forbidden bandwidth, eliminating network cross-talk and bounding routing overhead.
Lock	100 μs (derived)	Acquisition Timeout	The strict time limit for FUSE-layer atomic lock acquisitions ($T_2/16$).

Fractal Hilbert scaling delivers a 5.3x state volume advantage.



Condition:	N=12 nodes
AuraFS Replicas:	2
State Vol:	39.4
Euclidean State Vol:	12
Advantage:	~3.2x

Condition:	N=42 nodes
AuraFS Replicas:	2
Euclidean State Vol:	158
Advantage:	~3.7x

Condition:	N=100 nodes
AuraFS Replicas:	3
Euclidean State Vol:	530
Advantage:	100
	5.3x

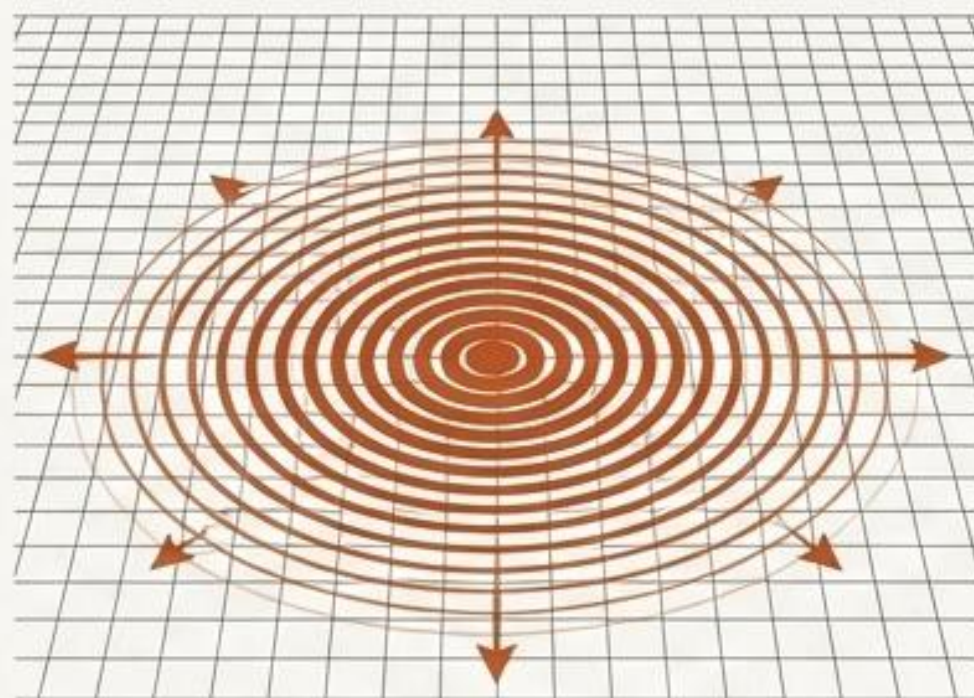
AuraFS achieves $D_{\text{eff}} \approx 3.02$ at $N=100$. By leveraging a fixed scaling bias of 5.3, the system requires only 3 replicas for 100 nodes, drastically reducing storage overhead without sacrificing availability.

Anderson Localization extends data coherence to 1600 μ s without active error correction.

Baseline Transmon:
100 μ s T_2 window.

Disordered Potential Lattice Landscape

Standard Euclidean Plane ($d_s > 2$) : Infinite Diffusion



Signal spreads uncontrollably,
leading to decoherence.

AuraFS Fractal Lattice ($d_s = 1.37$) : Localized Traps

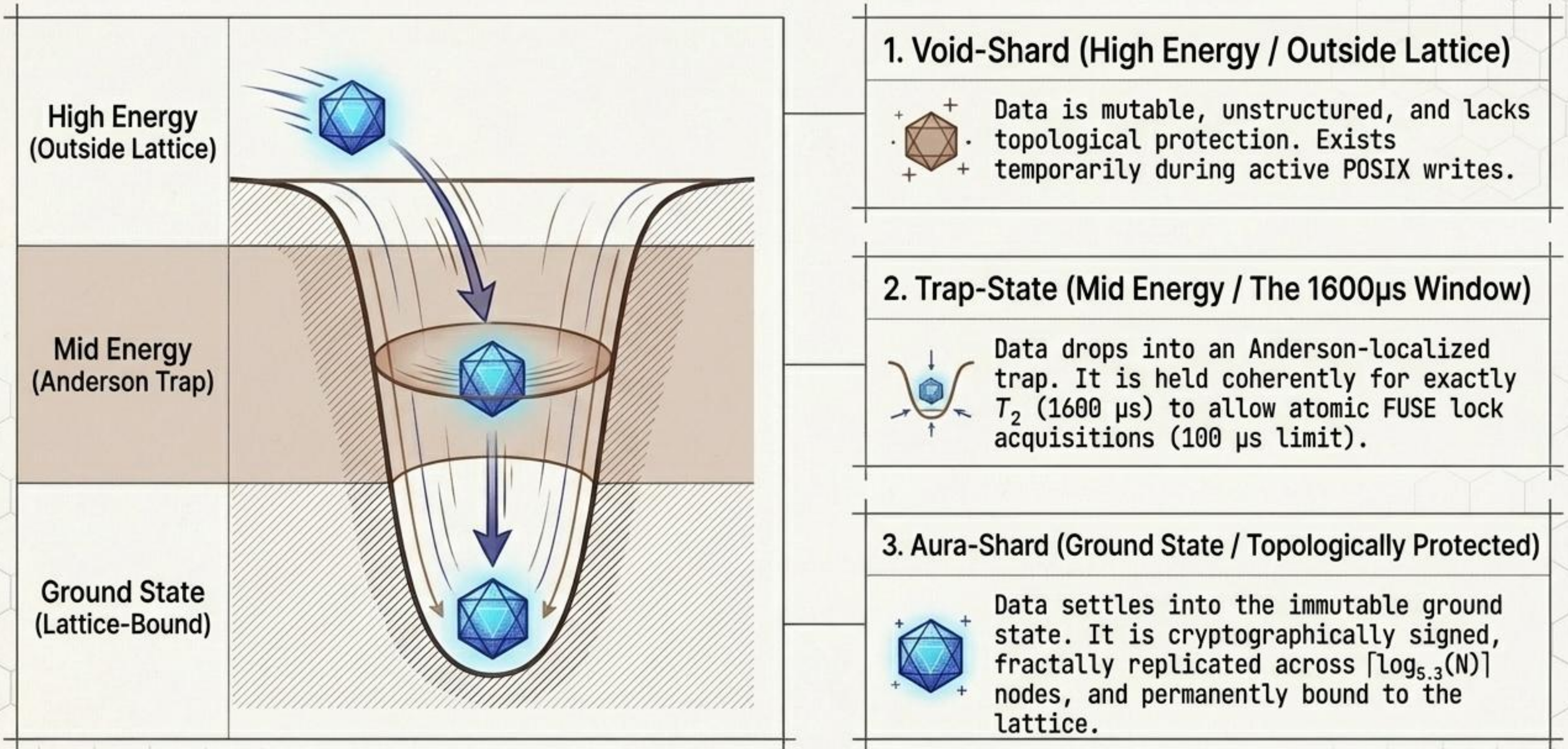


Waves become trapped in geometric
wells, preserving coherence.

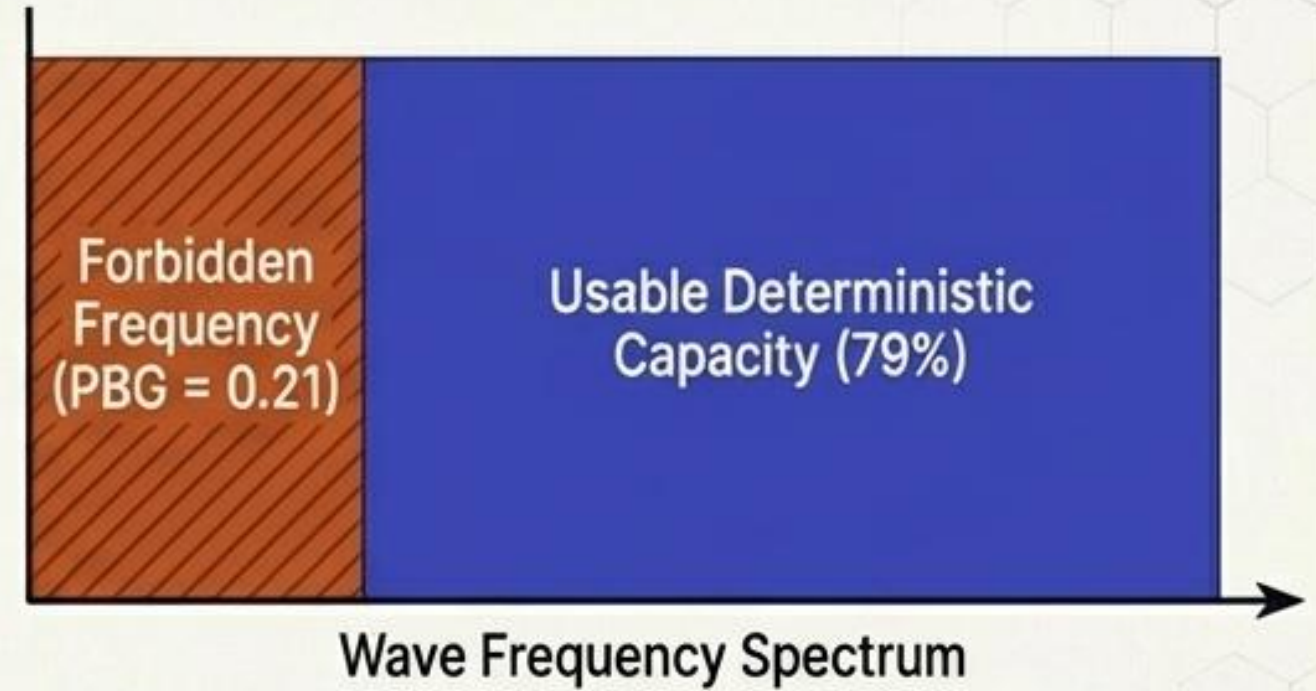
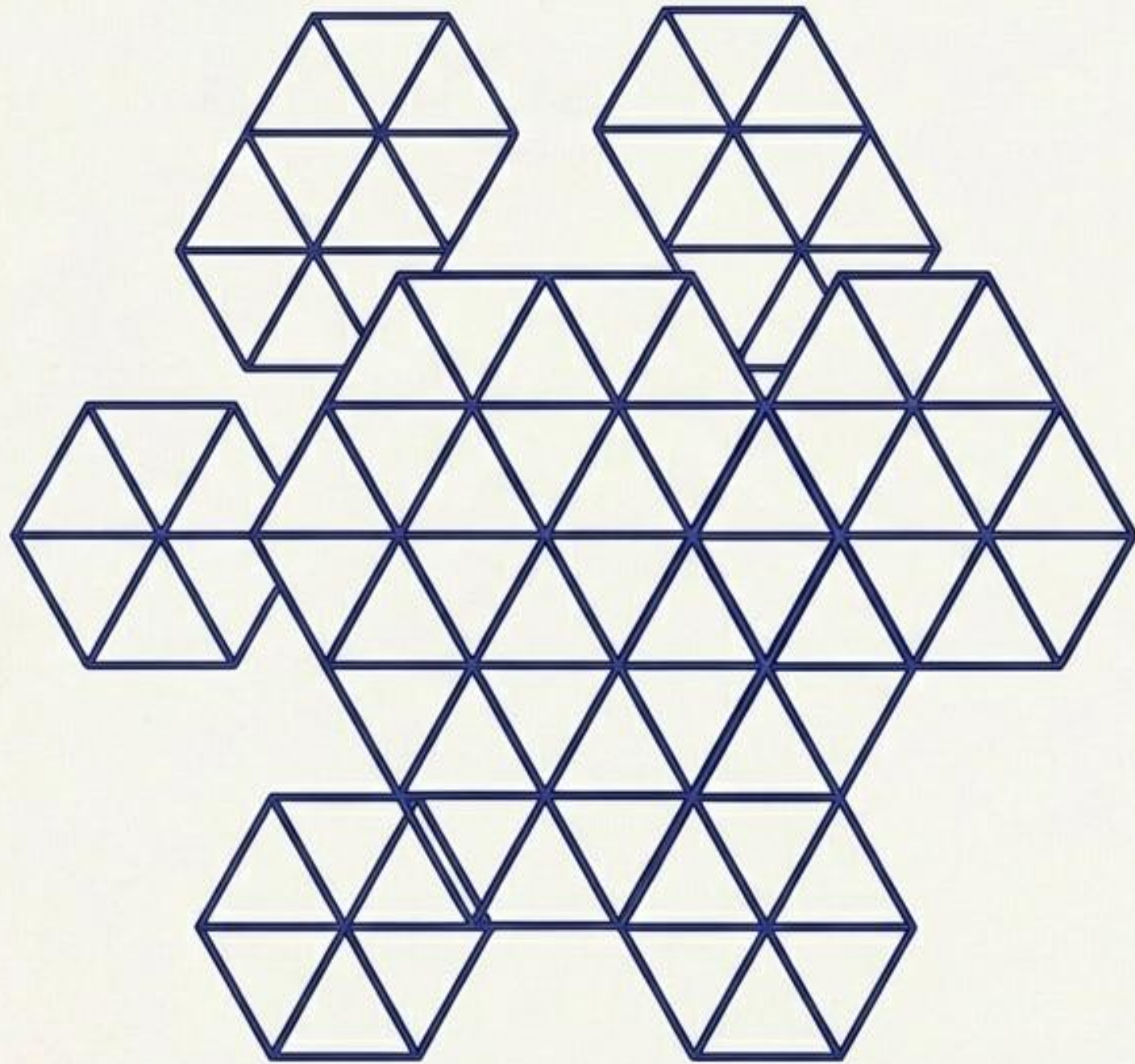
AuraFS Fractal Lattice:
1600 μ s T_2 window
(a 16x fidelity
improvement).

Mechanism: Inverse Participation Ratio (IPR)
confirms strong localization ($PR \approx 21.2$).
Because $d_s < 2$, eigenstates cannot diffuse, naturally
suppressing decoherence by $\gamma_{err}/\gamma_0 = 1/16$.

The physical lifecycle of an AuraFS data shard.



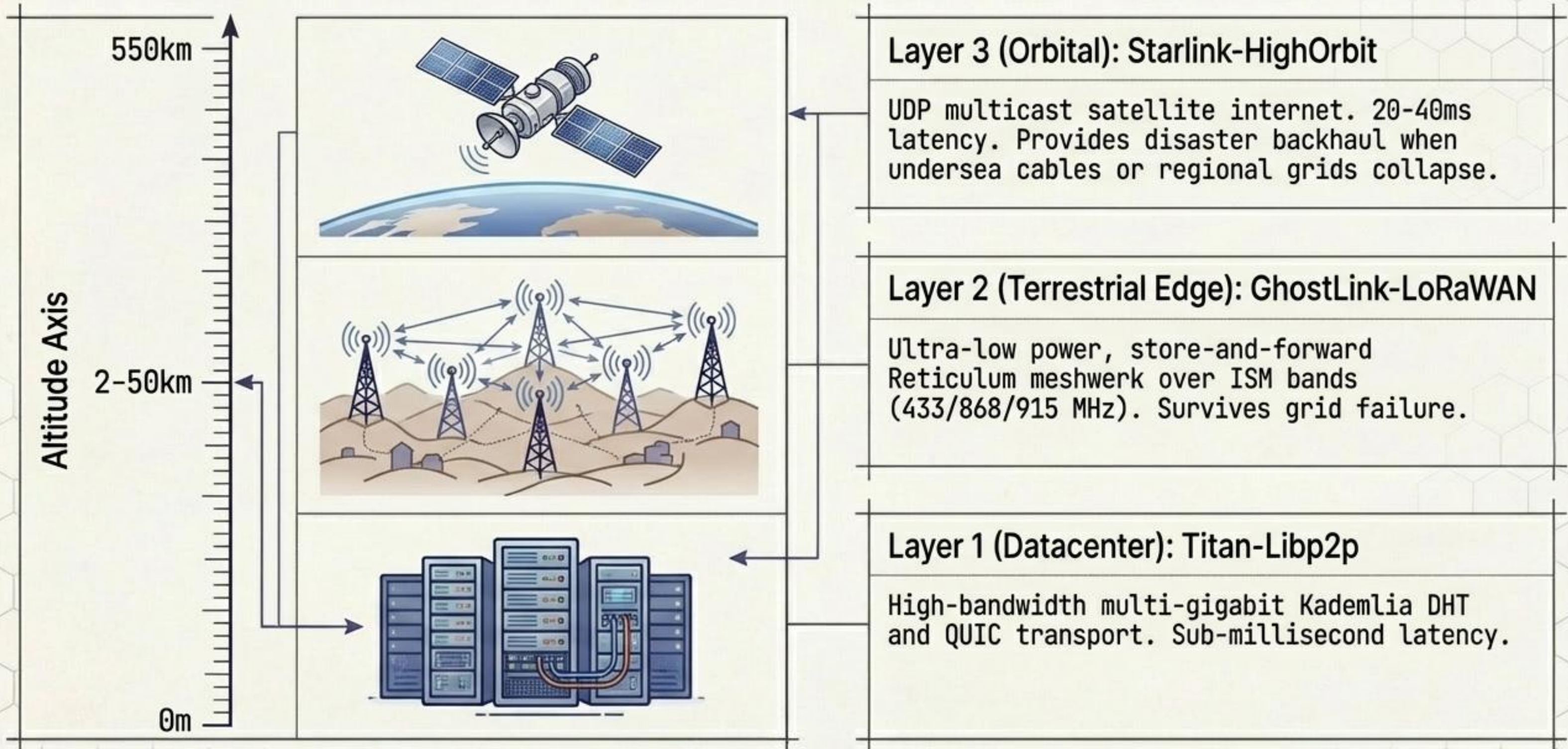
Photonic Band Gap constraints eliminate network congestion.



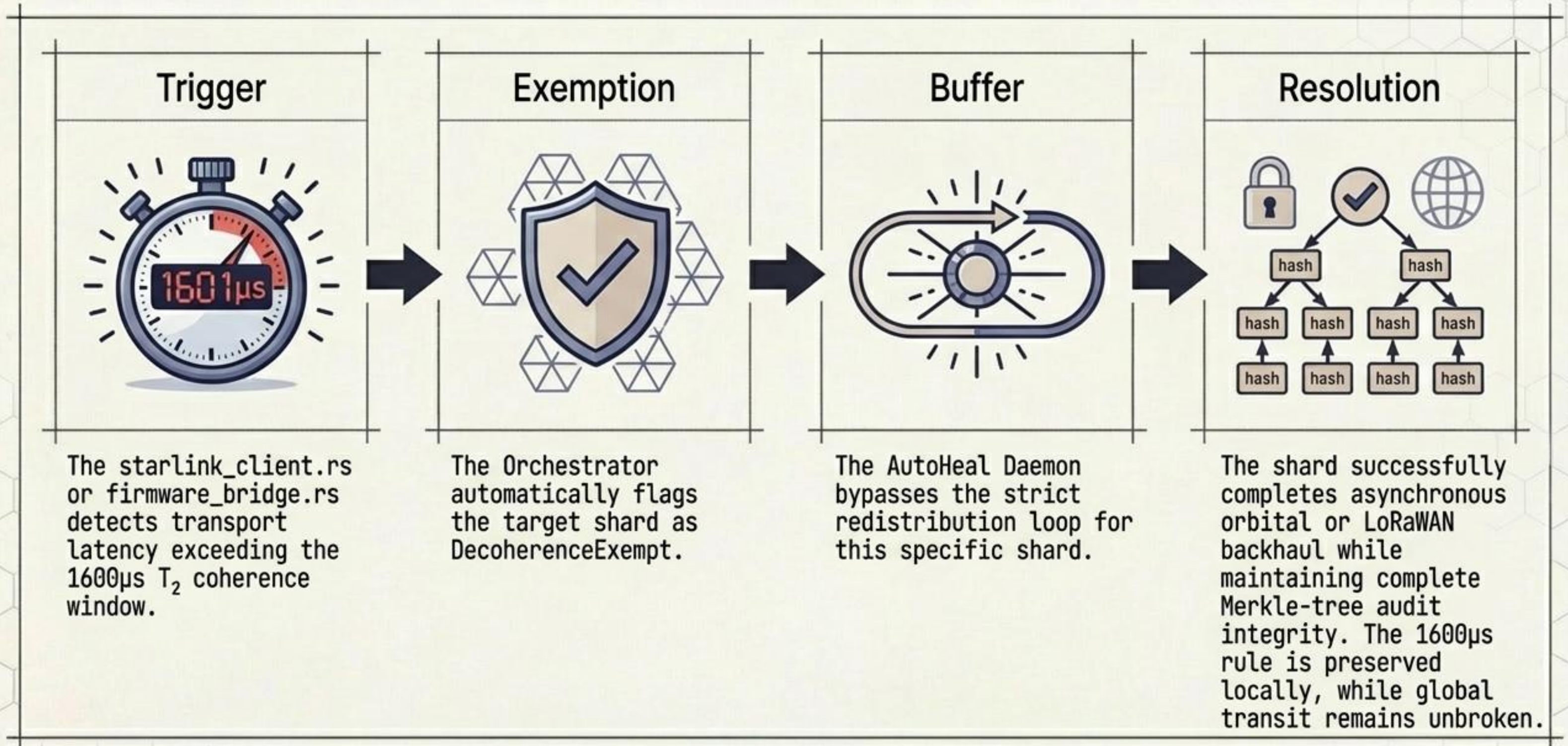
The meshwerk routing engine is bound by a 21% complete transverse magnetic (TM) band gap. Electromagnetic propagation is prohibited here, strictly eliminating cross-talk.

This physical constraint leaves exactly 79% usable capacity. Routing decisions become deterministic based on topology snapshots, completely bypassing the need for statistical load balancing algorithms.

The Triple-Network Topology ensures absolute operational survival.



Bridging the speed of light: Asynchronous Decoherence Exemptions.



NIST Level 5 Post-Quantum Cryptographic Substrate

Signatures (FIPS 204)

Dilithium-5 [Fully Implemented]

Quantum-signs all Holographic Logger entries and shard transitions strictly within the coherence window.

Key Encapsulation (FIPS 203)

Kyber-1024 [Phase 2 In-Progress]

ML-KEM integration to permanently replace legacy hybrid key exchanges.

Tree Hashing (FIPS 202)

SHA3-256 [Fully Implemented]

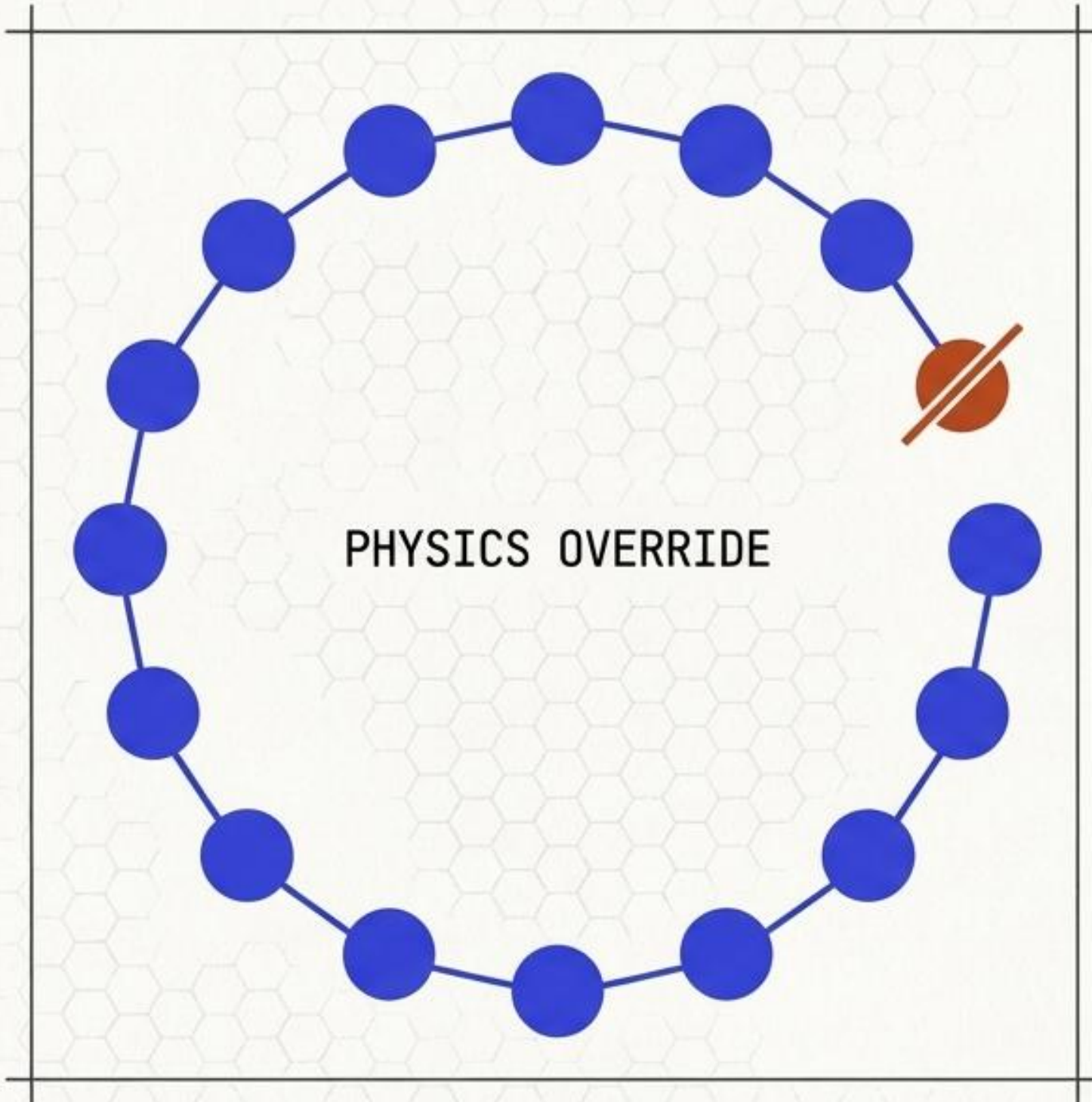
Secures the topological Merkle trees, providing immutable shard inclusion proofs.

At-Rest Encryption (FIPS 197)

AES-256-GCM [Fully Implemented]

Encrypts Dilithium private keys and secures local block stores with strict symmetric protection.

S.A.G.E.S. Protocol: Byzantine-Tolerant Physics Governance



Immutable Configuration

The physics constants (`aurafs.toml`) are not arbitrary tuning parameters; they are mathematically bound and structurally essential.

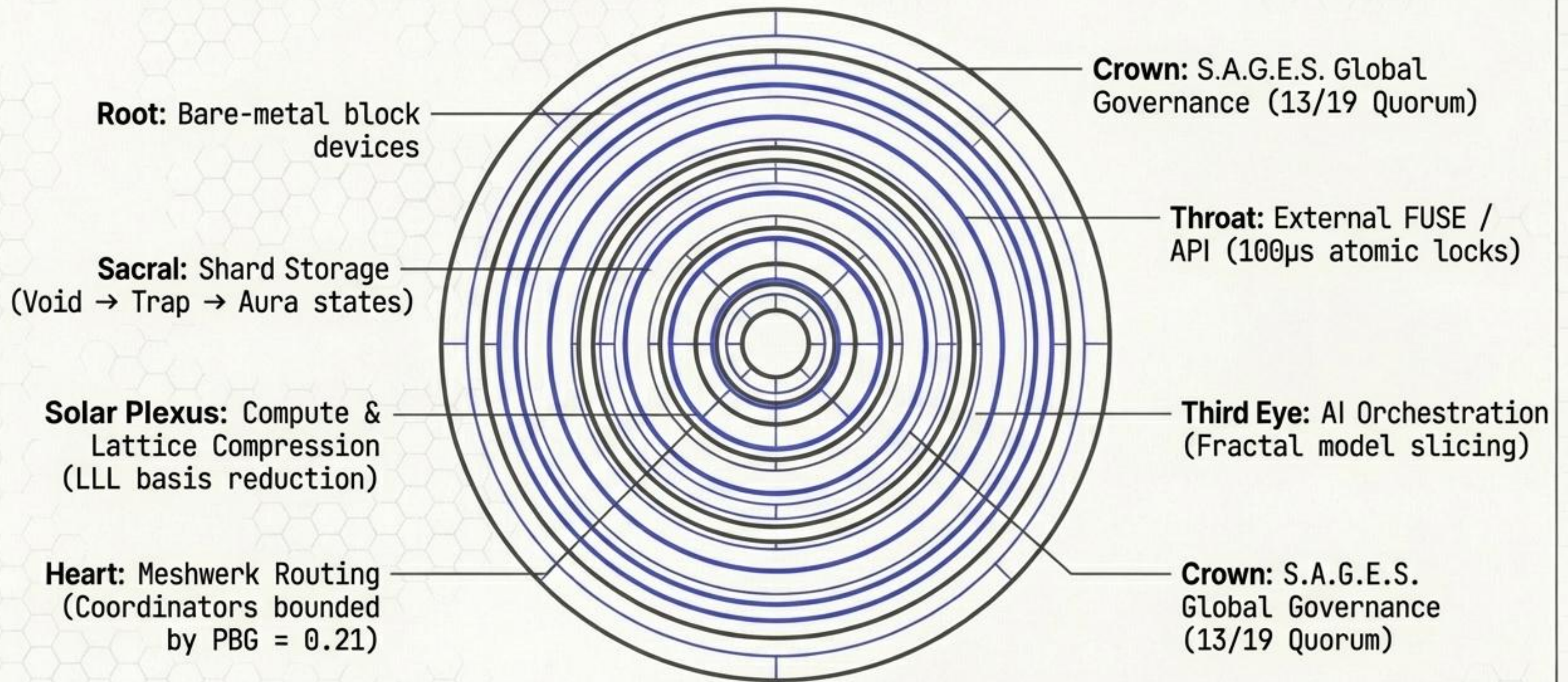
The Consensus Standard

Any modification to the fundamental physical constants requires a strict Byzantine $3f+1$ quorum to process.

Execution

Exactly 13 out of 19 Sentinel signatures are required to execute a PHYSICS OVERRIDE commit. Unilateral changes are mathematically and cryptographically impossible.

The Chakra DataCore: A 7-Zone Fractal Topology Synthesis



TRL-4 Codebase Validation: Zero Magic Numbers.

Phase 1 Audit Results: SUCCESS

Files Scanned: 462 Rust files

Violations Remediated: 31

Status: 100% of physical constants are now strictly bound to the `physics::INVARIANTS` singleton.

Action: `physics_audit.yml` passed

Coverage: $\geq 90\%$ in core physics module.

Code Remediation

Before (Magic Number Violation)

```
let replica_count =  
(nodes).log(5.3).ceil();
```

After (Strict Physics Binding)

```
let replica_count = (nodes).log  
(INVAVARIANTS.hilbert_scaling_bias)  
.ceil();
```


Deployment Strategy: Phase 2 Hardening & Launch.

